Overcoming Operational Bottlenecks In Laundry Services: A Case Of Uclean

Mid Term Submission for the BDM Capstone Project

Submitted By

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1. <u>Executive Summary:</u>

This BDM capstone project, titled *Operational Bottleneck Analysis and Optimization of Laundry Services at UClean*, focuses on a franchise located in Sector 37D, Gurgaon. UClean, India's leading B2C laundry and dry-cleaning chain, operates on a franchise model and offers services such as washing, dry-cleaning, steam ironing, and doorstep delivery. The aim of this study is to identify operational inefficiencies and provide data-driven solutions to improve service quality, customer satisfaction, and business profitability.

An in-depth review of billing and operational data over six months revealed consistent issues of long turnaround times caused by poor task allocation and staff imbalances. Orders were frequently delayed, especially during weekends, leading to dissatisfaction and drop in repeat customer visits. High cancellation rates were also observed, often triggered by miscommunication and service delays, resulting in underutilized resources and lost revenue. In addition, the franchise struggled with fluctuating seasonal demand due to lack of forecasting tools, leading to resource overload during peak times and inefficiencies during off-peak periods.

To tackle these challenges, the project employs data analysis techniques such as turnaround time tracking, cancellation pattern identification, and time-series forecasting for demand planning. Tools like Excel and Python were used for data cleaning, visualization, and trend detection. These analyses help map inefficiencies, understand customer behavior, and project future demand more accurately.

The insights gained from this project aim to reduce order delays, minimize cancellations, and optimize resource allocation during seasonal spikes. As a result, the franchise can achieve higher customer satisfaction, improved service reliability, and enhanced operational sustainability. This data-driven approach provides a foundation for strategic decisions that will strengthen UClean's market position and long-term profitability.

2. Proof of Originality:

The data applied in this project is primary data, which is collected directly by interacting with the business owner through one-on-one interactions and by compiling details from physical receipts and owner interactions. Order dates, pickup and delivery times, cancellations, and revenue records were transcribed into Excel for analysis. This ensures originality, accuracy, and relevance to the identified problems of turnaround delays, cancellations, and seasonal demand forecasting.

Link to the photos, video, data, and other related files:

https://drive.google.com/drive/folders/12YpOrYlDa1A954LQ82jW_871ViGpl5k9?usp=sharing

Letter from the organization:

https://drive.google.com/file/d/1ePoCq70JiuIPLWnodBD4_uL4GaIzhoB2/view?usp=sharing

3. Metadata:

Description:

The dataset for this project is compiled in a single Excel workbook titled **"UClean Laundry Data (Mid-Term Submission)"**, containing detailed operational records of **UClean's Sector 37D, Gurgaon franchise**. The data spans **April 2025 to July 2025**, providing a comprehensive four-month view of order activity, staff workload, and service performance. It was created to analyze turnaround time delays, order cancellations, and seasonal demand variations, the three major operational bottlenecks faced by the store.

Key Data Points:

• **1117 unique orders recorded.**

• **Service Range:** Wash & Fold, Wash & Iron, Premium Laundry, Dry Clean, Steam Iron, Others.

• **Order Value Range:** ₹150 - ₹1,500 per order (total billed amount based on item count and service type, measured per piece or per kilogram).

• **Order Status:** Completed / Cancelled, with specific reasons (*Late Pickup, Miscommunication, Customer Changed Mind*).

Operational Data:

The dataset contains **16 columns**, covering service type, staff allocation, pricing, timelines, and revenue. It provides a structured framework to assess efficiency, customer satisfaction, and workload distribution.

• **Order_ID:** Unique identifier for each order; enables performance tracking and record linkage.

• **Order_Date:** Date when the order was placed; serves as the base for analyzing trends and seasonal demand.

• **Month:** Derived from Order_Date; used for monthly and seasonal demand comparison.

• **Service_Type:** Defines the type of laundry service requested; essential for workload segmentation and service-level analysis.

• **Pickup_Date:** Indicates when the order was collected from the customer; starting point for turnaround tracking.

• **Unit:** Specifies the unit of measurement (*piece* or *kg*); helps interpret pricing and workload distribution accurately.

• **Item_Count:** Number of items in each order; directly affects total order value and turnaround time.

• **Base_Price (₹):** Standard price per unit of service; used to calculate order-level revenue.

• **Total_Order_Value (₹):** Total amount billed per order (Base_Price × Item_Count); vital for financial performance evaluation.

• **Expected_Delivery_Date:** The promised completion date; used to compare against actual delivery for delay analysis.

• **Staff_ID:** Identifies the employee assigned to the order; useful for analyzing staff performance and efficiency.

• **Actual_Delivery_Date:** Actual date of service completion; compared with expected date to calculate delay.

• **Delay (Days):** Computed as Actual Delivery - Expected Delivery; measures service efficiency and time adherence.

• **Status:** Indicates whether the order was completed or cancelled; core metric for reliability and operational consistency.

• **Cancellation Reason:** Provides the cause of order cancellation; helps identify recurring operational or communication issues.

• **Refund:** Amount refunded (if any) for cancelled or delayed orders; reflects customer compensation and financial loss.

Each of these variables provides a critical layer of insight from operational performance and staff productivity to financial output and customer experience forming the foundation for the project's analytical focus on turnaround efficiency, cancellation management, and seasonal demand forecasting.

4. Descriptive Statistics:

This section presents the key insights derived from the descriptive statistics of UClean's operational dataset, highlighting revenue distribution, service demand, and sales performance trends from **April 2025 to July 2025**.

Year-Month	Total Revenue	Mean Revenue	Total Item sold
April, 2025	324997	136.4	2383
May, 2025	363695	130.16	2794
June, 2025	123423	134.74	916
July, 2025	136875	142.43	961

From April to July 2025, both **revenue** and **item sales volume** displayed noticeable seasonal variation. The months of **April and May 2025** recorded the **highest operational activity**, generating combined revenue exceeding **₹6.8 lakh** and accounting for nearly **65% of total items processed**. This demand surge aligns with the **wedding and summer cleaning season**, when customers require more frequent laundry and dry-cleaning services.

However, a sharp decline is observed in **June and July 2025**, where revenue dropped to nearly **half of the previous months' figures**, reflecting an **off-season slump** in demand. Average order revenue remained relatively stable, suggesting consistent pricing and order composition even as total volume fluctuated.

Service Type	Total Order Value	Total Item Count	Avg Order Value
Dry Clean	179768	782	229.88
Others	159101	718	221.59
Premium Laundry	259209	1041	249
Steam Iron	235792	3187	73.98
Wash & Fold	232056	2344	99
Wash & Iron	226292	1628	138.99

Among the six service categories, **Premium Laundry** emerged as the **top revenue contributor** with a total order value of ₹2,59,209 and an **average order value of ₹249**, indicating its premium customer base and higher per-order profitability. **Steam Iron** and **Wash & Fold**, despite having lower ticket sizes (₹73.98 and ₹99 respectively), generated substantial total revenue due to **high order volumes**, suggesting they are the **most frequently availed services**.

Dry Cleaning and **Wash & Iron** also performed consistently, catering primarily to working professionals who prefer convenience-oriented services. The **'Others'** category, which includes special or one-time services, contributed modestly to revenue but showed potential for targeted marketing and expansion.

Key Insights

1. **Peak Demand Period:** April-May 2025 showed strong customer demand, aligning with seasonal factors, while June-July reflected a clear off-season pattern.
2. **Revenue Dependence on Service Mix:** Premium Laundry contributed the highest per-order revenue, whereas Steam Iron drove customer frequency.
3. **Stable Pricing:** Mean revenue per order remained within a narrow range, implying standardized pricing and consistent order size across months.
4. **Operational Implications:** The data reinforces UClean's need for **seasonal resource planning**, including staff scheduling and machine utilization optimization, to handle fluctuations efficiently.

5. <u>Detailed Explanation of Analysis Process/Method:</u>

5.1 Data Loading and Initial Exploration

The dataset titled **“UClean Laundry Data (Mid-Term Submission)”** was analyzed using **Microsoft Excel**. It contains **1117 order-level records** from **April to July 2025** for UClean’s Sector 37D, Gurgaon franchise. Key variables include order timelines, service types, staff assignment, order value, delay days, order status, and cancellation reasons. Initial exploration involved checking data completeness, understanding variable distributions, and calculating total orders, total revenue, and monthly order volumes to gain a preliminary view of operations.

5.2 Data Preprocessing

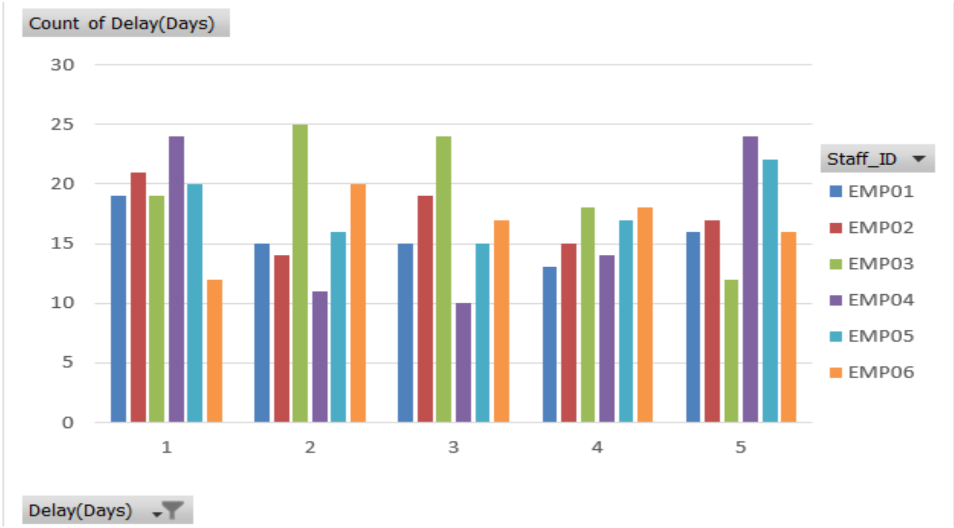
To ensure data accuracy, all date fields were standardized, and derived variables such as **Month** and **Delay (Days)** were created. Cancelled orders were clearly identified by missing actual delivery dates. Price-related columns were cleaned by removing units and currency symbols for numerical consistency. These preprocessing steps were essential to avoid incorrect calculations and enable reliable pivot-based analysis.

5.3 Descriptive Statistics

Descriptive statistics were generated to summarize operational performance. Monthly revenue, average order value, and total item counts were calculated to understand service demand and workload intensity. Service-wise aggregation highlighted high-volume services such as **Steam Iron** and **Wash & Fold**, while **Premium Laundry** showed the highest average order value. These statistics established a quantitative base for further analysis.

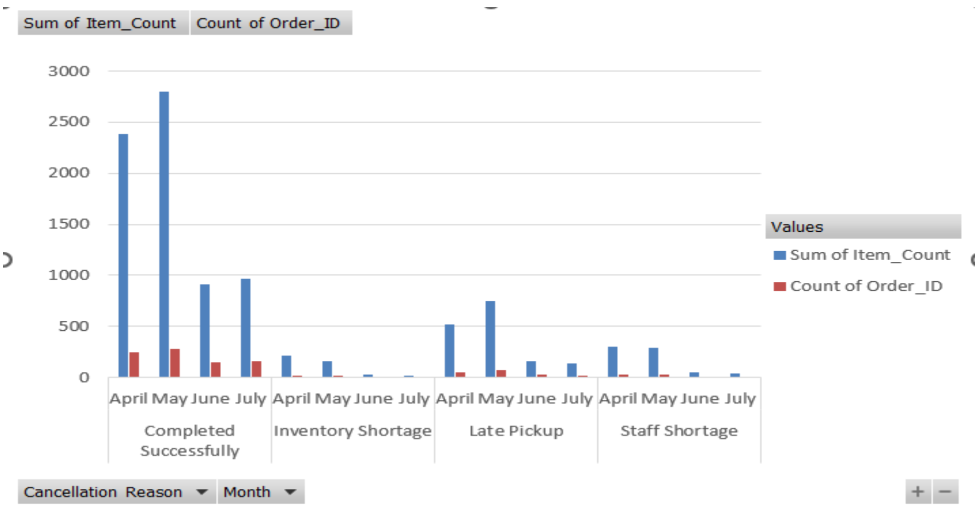
5.4 Turnaround Time and Delay Analysis

Turnaround delays were analyzed using **Delay (Days)** and **Staff ID** through frequency-based Pivot Tables and charts. This method was chosen over simple averages as it highlights how often delays occur and identifies employee-level workload imbalance. The analysis revealed frequent 2-3 day delays, confirming inefficiencies in task allocation and capacity utilization.



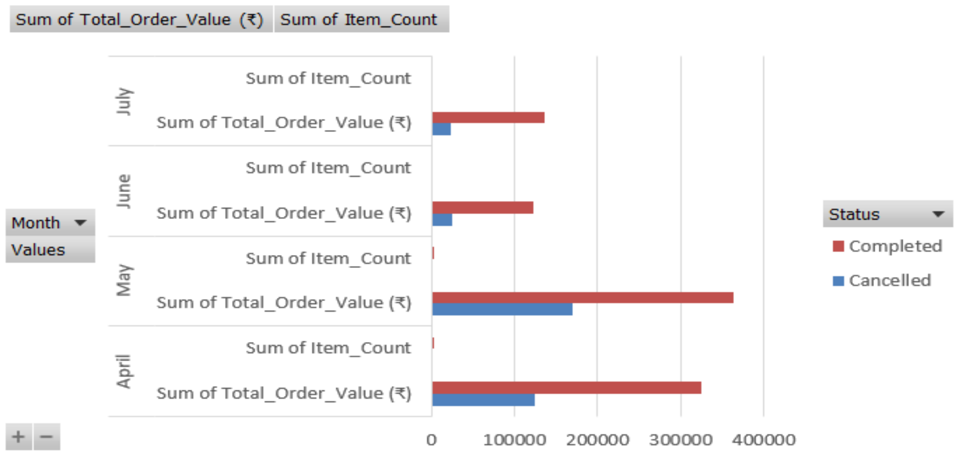
5.5 Cancellation Analysis

Cancellations were examined using **Order Status**, **Cancellation Reason**, and **Month**. Pivot analysis showed higher cancellation volumes in **April and May**, mainly due to late pickup and staff shortages. This confirms that operational stress during peak demand periods leads to service disruptions.



5.6 Seasonal Demand Fluctuation Analysis

Monthly aggregation of **order status, item count, and order value** was used to study seasonal trends. Demand peaked in **April-May** and declined sharply in **June-July**, indicating significant seasonal fluctuation. This explains the rise in delays and cancellations during high-demand months.



5.7 Summary of Findings

The analysis shows that **long turnaround times**, **high cancellations**, and **seasonal demand fluctuations** are strongly interlinked. Peak-season demand overwhelms operational capacity, leading to delays and service failures. These findings validate the problem statements and justify the need for forecasting-based resource planning and workflow optimization.

6. <u>Results and Findings:</u>

This section presents the key insights derived from the analysis of UClean's operational data, focusing on service delays, order cancellations, and seasonal demand fluctuations. The findings are based on pivot tables and charts generated from primary data.

6.1 Service-wise Revenue Distribution

Service-wise revenue analysis shows a fairly balanced contribution across services, with Premium Laundry, Steam Iron, and Wash & Iron emerging as major contributors. In April and May, premium and time-sensitive services contributed a higher share of total revenue, reflecting increased customer demand during peak months.

In contrast, June and July show a shift toward basic services such as Wash & Iron, indicating a reduction in premium demand during off-peak periods. This highlights changing customer preferences across seasons.

6.2 Turnaround Time and Delay Patterns

Delay analysis reveals that 2-3 day delays are the most frequent, followed by 4-5 day delays. The recurrence of moderate delays indicates that service inefficiencies are systematic rather than isolated.

Employee-wise comparison shows uneven distribution of delayed orders, suggesting workload imbalance and capacity constraints during high-demand periods.

6.3 Order Cancellations and Disruptions

Cancellation analysis indicates significantly higher cancellation volumes in April and May, primarily due to Late Pickup and Staff Shortage. These cancellations often involve larger item counts, resulting in both revenue loss and wasted operational effort.

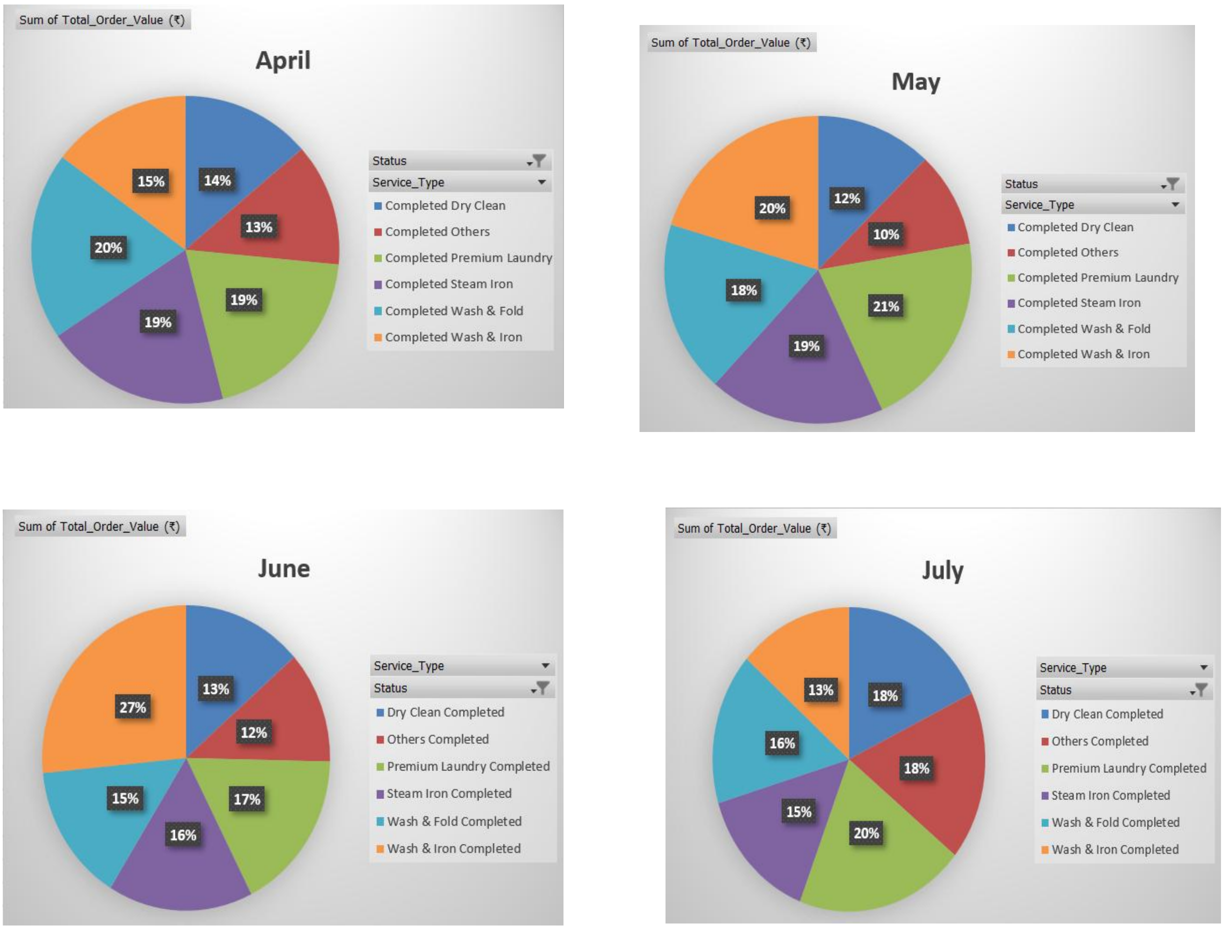
Cancellation rates drop noticeably in June and July, confirming that cancellations are closely linked to demand pressure rather than customer behavior alone.

6.4 Seasonal Demand Fluctuation

Monthly order value and item count trends show a clear demand peak in April-May, followed by a sharp decline in June-July. This demand volatility explains the rise in delays and cancellations during peak months and underutilization during off-peak periods.

6.5 Overall Insights

The analysis confirms that seasonal demand surges lead to operational overload, resulting in service delays and order cancellations. These findings validate the identified problem statements and highlight the need for better demand forecasting, workforce planning, and workflow management to improve service efficiency and customer satisfaction.



Service Type Monthly Performance and Seasonal Analysis

7. Conclusion:

This study analyzed turnaround delays, order cancellations, and seasonal demand patterns at UClean from April to July. The analysis revealed frequent service delays due to uneven workload distribution and higher cancellation rates during peak months, primarily caused by late pickups and staff shortages. Seasonal demand trends showed a surge in April-May followed by reduced demand in June-July, leading to operational imbalance. Strengthening demand forecasting, staff planning, and workflow management can help reduce delays, control cancellations, and improve overall service efficiency.